# Statistical Analysis of Economic Growth and Its Determinants in India

**Data Source:** World Bank World Development Indicators (WDI)

**Country:** India (Code: IND)

**Analysis Period:** 1991–2025 (35 annual observations)

**Report Generated:** 2026-10-08T11:18:27.144038+00:00

## Executive Summary & Key Findings

• Over the period 1991–2025 (35 observations), India's average annual GDP Growth was 6.14% (range: -5.78% to 9.69%).

• Multiple OLS regression accounts for 28.37% of the variance in GDP Growth (R2 = 0.2837, F = 2.2969, p = 0.0710).

• No individual regressors are statistically significant at the strict 5% level in this specification.

• Diagnostic Checks: Durbin-Watson statistic (1.74) indicates no severe autocorrelation, while Breusch-Pagan test (p = 0.1222) confirms homoscedasticity.

## 1. Descriptive Statistics

Analysis of annual macroeconomic indicators for India over the period 1991–2025 (35 observations):

| Variable | Count | Mean | Median | Std Dev | Min | Max | Range | Q1 | Q3 | IQR | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **GDP_Growth** | 35 | 6.14 | 7.10 | 2.80 | -5.78 | 9.69 | 15.47 | 5.03 | 7.86 | 2.83 | -2.49 | 8.85 |
| **Inflation** | 35 | 6.89 | 6.33 | 3.11 | 2.40 | 13.87 | 11.47 | 4.27 | 9.23 | 4.96 | 0.69 | -0.60 |
| **FDI*Net*Inflows** | 35 | 1.26 | 1.06 | 0.80 | 0.03 | 3.62 | 3.59 | 0.67 | 1.74 | 1.07 | 0.75 | 0.74 |
| **Exports** | 35 | 17.64 | 19.16 | 5.41 | 8.49 | 25.43 | 16.94 | 12.01 | 22.16 | 10.16 | -0.36 | -1.35 |
| **Gross*Capital*Formation** | 35 | 32.10 | 32.12 | 5.17 | 22.72 | 41.95 | 19.23 | 29.04 | 34.85 | 5.81 | 0.00 | -0.80 |
| **Unemployment** | 35 | 7.18 | 7.61 | 1.07 | 4.17 | 7.86 | 3.69 | 7.56 | 7.64 | 0.08 | -2.30 | 3.91 |

## 2. Probability & Distribution Analysis

Evaluation of distribution shape and formal tests for normality:

| Variable | Shapiro-Wilk Stat | Shapiro-Wilk p | Anderson-Darling Stat | Skewness Type | Kurtosis Type | Recommendation |
|---|---|---|---|---|---|---|
| **GDP_Growth** | 0.7735 | 0.0000 | 1.9705 | Negatively Skewed (Left-tailed) | Leptokurtic (Heavy-tailed/Peaked) | Evidence of non-normality (α = 0.05) |
| **Inflation** | 0.9196 | 0.0139 | 1.0397 | Positively Skewed (Right-tailed) | Platykurtic (Light-tailed/Flat) | Evidence of non-normality (α = 0.05) |

| Variable | Shapiro-Wilk Stat | Shapiro-Wilk p | Anderson-Darling Stat | Skewness Type | Kurtosis Type | Recommendation |
|---|---|---|---|---|---|---|
| **FDI*Net*Inflows** | 0.9545 | 0.1554 | 0.4367 | Positively Skewed (Right-tailed) | Leptokurtic (Heavy-tailed/Peaked) | No evidence against normality (α = 0.05) |
| **Exports** | 0.9034 | 0.0049 | 1.2682 | Symmetric | Platykurtic (Light-tailed/Flat) | Evidence of non-normality (α = 0.05) |
| **Gross*Capital*Formation** | 0.9691 | 0.4193 | 0.3246 | Symmetric | Platykurtic (Light-tailed/Flat) | No evidence against normality (α = 0.05) |
| **Unemployment** | 0.5024 | 0.0000 | 8.2095 | Negatively Skewed (Left-tailed) | Leptokurtic (Heavy-tailed/Peaked) | Evidence of non-normality (α = 0.05) |

## 3. Correlation Analysis

Pairwise correlation analysis across 35 annual observations reveals linear (Pearson) and monotonic (Spearman) relationships among economic growth determinants in India. Associations with GDP Growth: Inflation: r = -0.1412 (p = 0.4184, Negative Very Weak / None); FDI*Net*Inflows: r = -0.1189 (p = 0.4962, Negative Very Weak / None); Exports: r = 0.1930 (p = 0.2665, Positive Very Weak / None); Gross*Capital*Formation: r = 0.3180 (p = 0.0626, Positive Weak); Unemployment: r = -0.2095 (p = 0.2270, Negative Weak).

### *Pairwise Relationships Table*

| Variable 1 | Variable 2 | Pearson r | Pearson p | Spearman rho | Spearman p | N Obs | Relationship | Significant (5%) |
|---|---|---|---|---|---|---|---|---|
| **GDP_Growth** | **Inflation** | -0.1412 | 0.4184 | -0.1331 | 0.4461 | 35 | Negative Very Weak / None | No |
| **GDP_Growth** | **FDI*Net*Inflows** | -0.1189 | 0.4962 | 0.1090 | 0.5332 | 35 | Negative Very Weak / None | No |
| **GDP_Growth** | **Exports** | 0.1930 | 0.2665 | 0.2308 | 0.1822 | 35 | Positive Very Weak / None | No |
| **GDP_Growth** | **Gross*Capital*Formation** | 0.3180 | 0.0626 | 0.4305 | 0.0098 | 35 | Positive Weak | No |
| **GDP_Growth** | **Unemployment** | -0.2095 | 0.2270 | -0.3751 | 0.0264 | 35 | Negative Weak | No |
| **Inflation** | **FDI*Net*Inflows** | -0.0940 | 0.5911 | -0.0950 | 0.5874 | 35 | Negative Very Weak / None | No |
| **Inflation** | **Exports** | -0.2212 | 0.2016 | -0.0401 | 0.8193 | 35 | Negative Weak | No |
| **Inflation** | **Gross*Capital*Formation** | -0.0957 | 0.5844 | -0.0717 | 0.6823 | 35 | Negative Very Weak / None | No |
| **Inflation** | **Unemployment** | 0.2869 | 0.0947 | 0.2320 | 0.1800 | 35 | Positive Weak | No |
| **FDI*Net*Inflows** | **Exports** | 0.6956 | 0.0000 | 0.6787 | 0.0000 | 35 | Positive Strong | Yes |
| **FDI*Net*Inflows** | **Gross*Capital*Formation** | 0.6248 | 0.0001 | 0.6118 | 0.0001 | 35 | Positive Strong | Yes |
| **FDI*Net*Inflows** | **Unemployment** | 0.1113 | 0.5246 | 0.1182 | 0.4988 | 35 | Positive Very Weak / None | No |
| **Exports** | **Gross*Capital*Formation** | 0.8249 | 0.0000 | 0.8255 | 0.0000 | 35 | Positive Very Strong | Yes |
| **Exports** | **Unemployment** | -0.3506 | 0.0389 | -0.0826 | 0.6369 | 35 | Negative Weak | Yes |
| **Gross*Capital*Formation** | **Unemployment** | -0.1558 | 0.3713 | -0.1399 | 0.4227 | 35 | Negative Very Weak / None | No |

## 4. Multiple Linear Regression Model (OLS)

**Dependent Variable:** `GDP_Growth`

**Independent Variables:** `Inflation, FDINetInflows, Exports, GrossCapitalFormation, Unemployment`

- **R-squared:** 0.2837
- **Adjusted R-squared:** 0.1602
- **F-statistic:** 2.2969 (p-value = 0.0710)
- **Residual Standard Error:** 2.5695
- **Sample Size (N):** 35 complete observations

### *Regression Coefficients Table (HC3 Robust SE)*

| Variable | Coefficient | HC3 Robust SE | t-statistic | p-value | 95% CI Lower | 95% CI Upper | Significant (5%) |
|---|---|---|---|---|---|---|---|
| **const** | -1.468261 | 5.243594 | -0.2800 | 0.7815 | -12.1926 | 9.2561 | No |
| **Inflation** | -0.111795 | 0.149001 | -0.7503 | 0.4591 | -0.4165 | 0.1929 | No |
| **FDINetInflows** | -1.862890 | 1.285579 | -1.4491 | 0.1580 | -4.4922 | 0.7664 | No |
| **Exports** | 0.009283 | 0.149119 | 0.0623 | 0.9508 | -0.2957 | 0.3143 | No |
| **GrossCapitalFormation** | 0.336227 | 0.216727 | 1.5514 | 0.1317 | -0.1070 | 0.7795 | No |
| **Unemployment** | -0.031613 | 0.334334 | -0.0946 | 0.9253 | -0.7154 | 0.6522 | No |

## 5. Regression Diagnostics

Econometric verification of OLS assumptions:

### *Diagnostic Tests Summary*

| Test | Statistic | p-value | Passed (5%) | Conclusion |
|---|---|---|---|---|
| **Durbin-Watson Autocorrelation** | 1.7440 | N/A | No strong evidence | DW = 1.744; the statistic is reasonably close to 2.0, suggesting no strong evidence of first-order autocorrelation. |
| **Breusch-Pagan Heteroscedasticity** | 8.6870 | 0.1222 | Fail to reject H■ | No statistically significant evidence of heteroskedasticity at α = 0.05. |
| **Shapiro-Wilk Residual Normality** | 0.9001 | 0.0040 | Reject H■ | Evidence of non-normality in residuals at α = 0.05. |
| **Cook's Distance (Influential Points)** | 0.6130 | N/A | 2 Exceeding Obs. | 2 observations (1991, 2020) exceed the Cook's D screening threshold (4/n ≈ 0.114). |

### *Variance Inflation Factors (VIF)*

| Predictor Variable | VIF | Multicollinear Warning | Status |
|---|---|---|---|
| **Inflation** | 1.1243 | No | Low Multicollinearity (VIF <= 5) |
| **FDINetInflows** | 2.6905 | No | Low Multicollinearity (VIF <= 5) |
| **Exports** | 5.7055 | Yes | Moderate Multicollinearity (VIF > 5) |
| **GrossCapitalFormation** | 3.3786 | No | Low Multicollinearity (VIF <= 5) |
| **Unemployment** | 1.7242 | No | Low Multicollinearity (VIF <= 5) |

## 6. Discussion & Methodological Cautions

***Caution:*** *Association vs. Causation: Neither Pearson/Spearman correlation coefficients nor Ordinary Least Squares (OLS) regression parameters demonstrate causal mechanisms. Observed relationships indicate statistical association only.*

***Caution:*** *Sample Size & Observational Data: Analysis is based on 35 annual time-series observations (1991–2025) for India. Time-series data can be subject to structural breaks, macro shocks (e.g., global recessions), and non-stationarity.*

***Caution:*** *Model Specification & Omitted Variables: OLS regression results reflect the 5 specified indicators (Inflation, FDI Net Inflows, Exports, Gross Capital Formation, Unemployment). Omitted macroeconomic variables (e.g., fiscal policy, monetary stance, institutional changes) may influence the estimated coefficients.*